Vision–language–action (VLA) models integrate visual perception, language understanding, and action generation across manipulation tasks and robot embodiments (Brohan et al., 2023; Kim et al., 2024; Black et al., 2025b; Wang et al., 2026b; 2025). Learning instruction-conditioned behaviors from demonstrations provides a unified mapping from multimodal inputs to robot actions. Extending VLA learning to humanoids could broaden task coverage in human environments, where locomotion, posture, and manipulation must be coordinated (Bjorck et al., 2025; Wei et al., 2026a; Hu et al., 2026). For example, transferring objects between tables requires grasping, turning, walking, and placement while maintaining balance (Fig. 1, bottom). Such tasks require grounding task intent and visual feedback in coherent whole-body actions, motivating the extension of robot policy architectures to humanoid loco-manipulation.

Humanoid policy learning commonly extends the backbone–action expert paradigm from manipulation to whole-body control. Pretrained vision–language models encode instruction and observation semantics (Fig. 2(a)) (Bjorck et al., 2025; Wei et al., 2026a; Hu et al., 2026), while video generators supply spatiotemporal priors for scene evolution (Fig. 2(b)) (Zheng et al., 2026; Li et al., 2026). These representations condition action generation across root, body, and hand motion. Unified predictors generate whole-body sequences (Fig. 2(d)) (Hu et al., 2026; Zheng et al., 2026); hybrid designs combine upper-body joint targets with lower-body locomotion commands executed by dedicated controllers (Fig. 2(e)) (Zhong et al., 2025; Jiang et al., 2025). The former captures cross-component dependencies; the latter couples upper-body control with learned locomotion and stabilization. Both transfer pretrained semantic and visual dynamics knowledge to humanoid loco-manipulation.

Despite these advances, two challenges remain. ❶ **Insufficient Whole-Body Motion Priors.** Compared with fixed-base manipulation, humanoid control requires higher-dimensional inter-joint and temporal coordination. Backbone pretraining (Bjorck et al., 2025; Wei et al., 2026a; Zheng et al., 2026; Li et al., 2026) does not directly equip action generators with priors aligned with target humanoid kinematics, leaving them to learn these priors, often from limited task-specific demonstrations. ❷ **Insufficient Differentiation of Motion Components.** Root, body, and hand control differ in scale and precision. Unified prediction (Hu et al., 2026; Zheng et al., 2026) may create competing objectives; hybrid execution (Zhong et al., 2025; Jiang et al., 2025) does not itself provide dedicated root, body, and hand generators. Both must accommodate component-specific requirements while maintaining whole-body coordination.